

• $|n\rangle \leftrightarrow \vec{V}$

* Hooke's Law:

(small motion)

$$\vec{F}_s = -kx$$



Spring constant

* $\vec{F}_s = -kx = m\vec{a} = m \frac{d^2x}{dt^2}$

↳ $m \frac{d^2x}{dt^2} + kx = 0 \Rightarrow \ddot{x} + \frac{k}{m}x = 0$

$$\frac{d^2x}{dt^2} + \left(\frac{k}{m}\right)x = 0$$



→ $\frac{d^2x}{dt^2} = -\left(\frac{k}{m}\right)x \Rightarrow$

$$x(t) = A \cos(\Omega t) + B \sin(\Omega t)$$

$$\Omega^2 = \frac{k}{m}$$

likewise: $\frac{d^2x}{dt^2} = +\left(\frac{k}{m}\right)x \Rightarrow$

$$x(t) = A \cosh(\Omega t) + B \sinh(\Omega t)$$

$$\Omega^2 = \frac{k}{m}$$

$$\begin{cases} \cosh(\theta) = \frac{e^\theta + e^{-\theta}}{2} \\ \sinh(\theta) = \frac{e^\theta - e^{-\theta}}{2} \end{cases}$$

$$\begin{aligned} e^{i\theta} &= \cos(\theta) + i \sin(\theta) \\ \cos(\theta) &= \frac{e^{i\theta} + e^{-i\theta}}{2} \\ \sin(\theta) &= \frac{e^{i\theta} - e^{-i\theta}}{2i} \end{aligned}$$

• $x(t) \sim e^{i\lambda t} \mid \frac{dx}{dt} = (i\lambda)x$

$$(i\lambda)^2 e^{i\lambda t} + (k/m)^2 e^{i\lambda t} = 0$$

$$-\lambda^2 + (k/m) = 0 \Rightarrow \lambda^2 = \frac{k}{m}$$

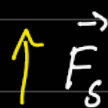
$\Omega^2 = \frac{k}{m}$

Solⁿ Simple Harmonic motion:

$$x(t) = A \cos(\Omega t) + B \sin(\Omega t)$$

ex 1

$\Delta \cdot t$



* Spring force always acts in the opposite direction
i.e. always pulling it back.

$$\sum_i (F_{ex})_i = F_s - F_g = m \frac{d^2 y}{dt^2} \quad \left| \begin{array}{l} y(0) = h_0 \\ \dot{y}(0) = v_0 \end{array} \right.$$

$$\downarrow \quad \downarrow$$

$$ky - mg = m \frac{d^2 y}{dt^2}$$

$$\frac{d^2 y}{dt^2} = \frac{k}{m} y - g$$

* $u \equiv \frac{k}{m} y - g \Rightarrow$

$$\frac{d^2 u}{dt^2} = \frac{k}{m} \frac{d^2 y}{dt^2}$$

$$\frac{d^2 y}{dt^2} = \frac{m}{k} \frac{d^2 u}{dt^2}$$

$$\frac{m}{k} \frac{d^2 u}{dt^2} = u \Rightarrow \frac{d^2 u}{dt^2} = \frac{k}{m} u$$

$$\Omega^2 = \frac{k}{m}$$

$$u(t) = A \cosh(\Omega t) + B \sinh(\Omega t)$$

$$\hookrightarrow = \frac{k}{m} y(t) - g$$

$$\Omega^2 y(t) - g = A \cosh(\Omega t) + B \sinh(\Omega t)$$

$$y(t) = \frac{1}{\Omega^2} (A \cosh(\Omega t) + B \sinh(\Omega t) + g)$$



* $u = \frac{k}{m} y - g$ only works when α, β is a constant

ex: $\frac{d^2 x}{dt^2} = \alpha x + \beta$

↑ ↑
Constants

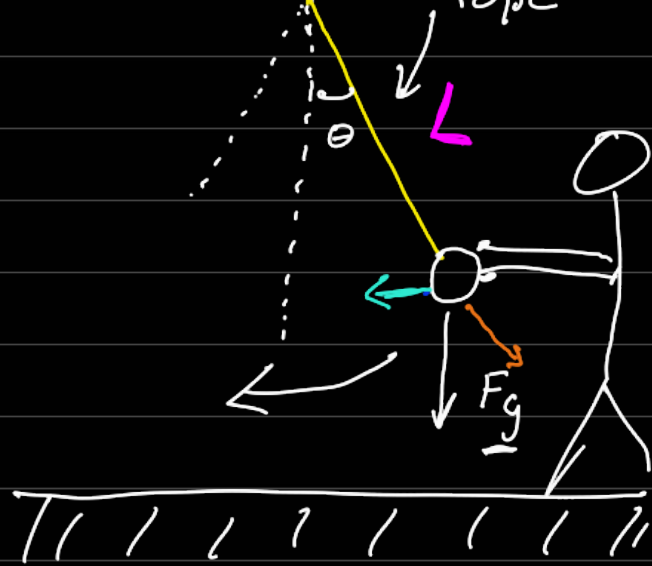
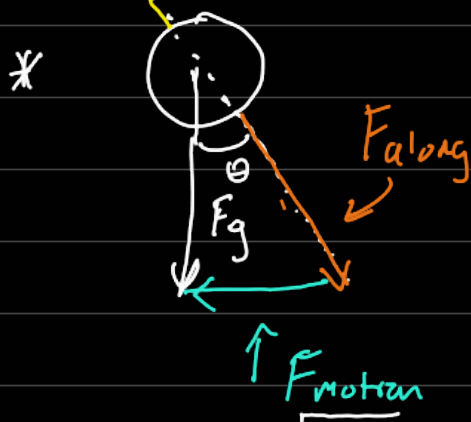
$$u = \alpha x + \beta$$

PS 5

* Above: zero friction $\rightarrow$ Undamped case

ring

① gravity pendulum:



$$\sin(\theta) = \frac{F_{\text{motion}}}{F_g}$$

$$F_{\text{motion}} = F_g \sin(\theta) = m \frac{d^2 x}{dt^2} \quad \leftarrow F_N$$



$$\sin(\theta(t)) = \frac{x(t)}{L}$$

$$x = L \sin(\theta)$$

$$\bullet \frac{dx}{dt} = L \cos(\theta) \frac{d\theta}{dt}$$

$$\bullet \frac{d^2 x}{dt^2} = L \left(\underbrace{-\sin(\theta) \frac{d\theta}{dt}}_{\frac{d \cos(\theta)}{dt}} \right) \cdot \frac{d\theta}{dt} + \cos(\theta) \left(\frac{d^2 \theta}{dt^2} \right)$$

$$= L \left(-\sin(\theta) \left(\frac{d\theta}{dt} \right)^2 + \cos(\theta) \frac{d^2 \theta}{dt^2} \right)$$

$$F_g \sin(\theta) = L \left(-\sin(\theta) \left(\frac{d\theta}{dt} \right)^2 + \cos(\theta) \frac{d^2 \theta}{dt^2} \right)$$

* Small angle approximation / Undamped

Undamped / Small angle

$$\theta \ll 1$$

$$\bullet \sin(\theta) \approx \theta$$

Small angle θ^2 or any term higher: we get

$$-\sin(\theta) \left(\frac{d\theta}{dt} \right)^2 \approx \theta (\dot{\theta}^2) \ll \ll 1$$

• Damping will come from terms that look similar

$$\Rightarrow \dots - \sin(\theta) \left(\frac{d\theta}{dt} \right)^2 \dots$$

We ignore $\sin(\theta) \approx \theta$ term
 In small angle $\cos(\theta) \approx 1 - \frac{\theta^2}{2}$ θ^2 : get rid of it

* left with $F_g \theta = L \ddot{\theta}$ | $F_g = m \frac{d^2 x}{dt^2}$

$$F_g = -mg$$

$$-mg \theta = (L \ddot{\theta}) m \Rightarrow \ddot{\theta} + \frac{g}{L} \theta = 0$$

* we solved this already

$$\textcircled{a} \theta(t) = A \cos(\omega t) + B \sin(\omega t)$$

$$\omega^2 = \frac{g}{L}$$

$\textcircled{b} (F_g)_{\text{Earth}} = -mg$ $\leftarrow g$: grav. acceleration (on Earth)

$$(F_g)_{\text{moon}} = -m \underbrace{\left(\frac{g}{6}\right)}_{\text{moon acceleration}} = -m \tilde{g}_{\text{moon}} \quad | \quad \tilde{g}_{\text{moon}} = \frac{g}{6}$$

$$\omega_{\text{moon}}^2 = \frac{\tilde{g}_{\text{moon}}}{L} = \frac{g}{6L} = \frac{1}{6} \omega_{\text{earth}}^2$$

$$\omega_{\text{moon}} = \frac{1}{\sqrt{6}} \omega_{\text{earth}}$$

